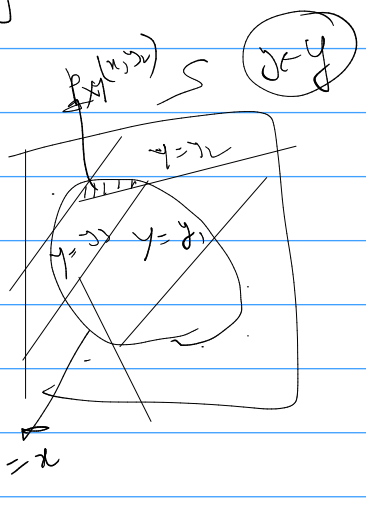
Sep 4 Bivariate distributions

Discrete: X, Y : two RVs "jointly distributed"

Outcome $\omega \longrightarrow (X(\omega), Y(\omega))$ $\quad x \in \mathcal{X}$, $y \in \mathcal{Y}$

Joint PMF: $p_{XY}(x,y) = P(X=x, Y=y)$ $\quad x \in \mathcal{X}, y \in \mathcal{Y}$

Marginal PMFs $\quad p_X(x) = P(X=x)$, $p_Y(y) = P(Y=y)$



event $(X=x) = \bigcup_{y \in \mathcal{Y}} (X=x \cap Y=y)$

$$p_X(x) = \sum_{y \in \mathcal{Y}} p_{XY}(x,y) \quad \text{"marginalization"}$$

$$p_Y(y) = \sum_{x \in \mathcal{X}} p_{XY}(x,y)$$

$$\sum_{x,y \in \mathcal{X} \times \mathcal{Y}} p_{XY}(x,y) = 1$$

$p_{XY}(x,y) \in [0,1]$

Ex: $X \in \{0,1\}$ $\quad Y \in \{0,1\}$

X \ Y	0	1
0	$p_{XY}(0,0)$	$p_{XY}(0,1)$
1	$p_{XY}(1,0)$	$p_{XY}(1,1)$

Y ← y ; X ↓ x

X \ Y	0	1
0	p_{00}	p_{01}
1	p_{10}	$p_{11} = 1 - p_{00} - p_{01} - p_{10}$

Marginals $\quad X \sim \{0, 1\}$ with $p_{00}+p_{01}$ and $p_{10}+p_{11} = 1-p_{00}-p_{01}$

$Y \sim \{0, 1\}$ with $p_{00}+p_{10}$ and $p_{01}+p_{11} = 1-p_{00}-p_{10}$

Ex:

x \ y	0	1
0	p	$\frac{1}{2}-p$
1	$\frac{1}{2}-p$	p

$p \in [0, \frac{1}{2}]$

$X \sim \{0, 1\}$ with $\frac{1}{2}, \frac{1}{2}$

$Y \sim \{0, 1\}$ with $\frac{1}{2}, \frac{1}{2}$

Ex:

P	ω_1	ω_2	ω_3	$X = \omega_1+\omega_2\omega_3$	$Y = \omega_2+\omega_3$
$\frac{1}{8}$	0	0	0	0	0
$\frac{1}{8}$	0	0	1	0	1
	0	1	0	0	1
	0	1	1	1	2
	1	0	0	1	0
	1	0	1	1	1
	1	1	0	1	1
$\frac{1}{8}$	1	1	1	2	2

X \ Y	0	1	2	
0	$\frac{1}{8}$	$\frac{2}{8}$	0	$\frac{3}{8}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
2	0	0	$\frac{1}{8}$	$\frac{1}{8}$
	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$	

Ex:

1% biased ($P(H)=0.4$) coins, 99% fair

Pick a coin at random

Toss it n times

$B = \{0, 1\}$ (fair, biased)

X = No. of heads

$$P_{BX}(0,k) = \frac{99}{100}\binom{n}{k}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k} \qquad k=0,1,\dots,n$$

$$P_{BX}(1,k) = \frac{1}{100}\binom{n}{k}(0.4)^k(0.6)^{n-k} \qquad k=0,1,\dots,n$$

Conditional PMFs

$P_X(x) = P(X=x)$

$P_Y(y) = P(Y=y)$

$P_{XY}(x,y) = P(X=x \cap Y=y)$

$$p_{Y|X}(y|x) \triangleq P(Y=y \mid X=x) = \frac{P(Y=y \cap X=x)}{P(X=x)} = \frac{p_{XY}(x,y)}{p_X(x)}$$

$$p_{XY}(x,y) = p_X(x) \cdot p_{Y|X}(y|x)$$

$$p_{XY}(x,y) = p_Y(y)\, p_{X|Y}(x|y)$$

Ex:

$X \backslash Y$	0	1	
0	p	$\frac{1}{2}-p$	$\frac{1}{2}$
1	$\frac{1}{2}-p$	p	$\frac{1}{2}$
	$\frac{1}{2}$	$\frac{1}{2}$	

$p \in (0, \frac{1}{2})$

$Y \sim \{0, 1\}$ with $\frac{1}{2}, \frac{1}{2}$

$Y|X=0 \sim \{0, 1\}$ with $2p, 1-2p$

$$p_{Y|X}(y|0) = \begin{cases} \frac{p_{XY}(0,0)}{p_X(0)} = 2p & y=0 \\ 1-2p & y=1 \end{cases}$$

$p_{Y|X}(y|0) \leftrightarrow p_{Y|X=0}(y)$

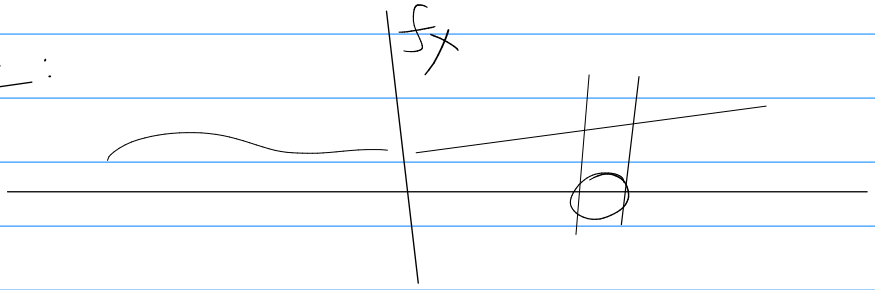
Ex: biased vs. fair coin ex above

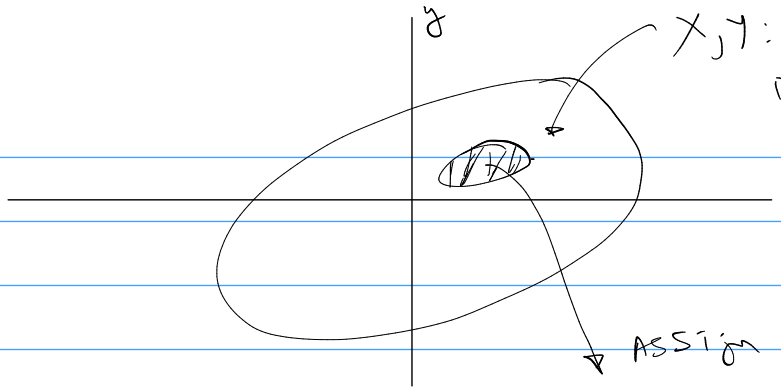
$X|B=0 \sim \text{Binom}(n, \frac{1}{2})$

$X|B=1 \sim \text{Binom}(n, 0.4)$

$B|X=k \sim \{0, 1\}$ (0 ↓ Bayes' thm, 1 ↓ Bayes' thm)

Continuous:





Joint density $f_{XY}(x,y)$

$f_{XY}: \mathbb{R}^2 \to \mathbb{R}$ $\quad f_{XY}(x,y) \geq 0$

$$\iint_{\mathbb{R}^2} f_{XY}(x,y)\,dx\,dy = 1$$

X, Y: jointly continuous.